A tetrahedrally symmetric four-component link whose three-component sublinks are Borromean

Neeraja Abhyankar

October 12, 2026

Abstract

We exhibit a link of four components in $\mathbb{R}^3$ with two properties: every one of its four three-component sublinks is the Borromean rings, and the embedding is carried onto itself by all twelve rotations of a regular tetrahedron. The components are given by twenty-four points with integer coordinates, generated from two of them by the symmetry group. We also prove two obstructions that explain why such a link must look complicated: no configuration of four *convex planar* components can have the required linking, and neither can any configuration in which each pair of components crosses only the minimum possible number of times.

1 Introduction

The Borromean rings are three closed loops, no two of which are linked, which nevertheless cannot be pulled apart; cut any one and the other two fall free. This paper is about a four-component analogue in which the Borromean property holds for every triple at once.

Problem 1.1. Find a link $L = K_1 \sqcup K_2 \sqcup K_3 \sqcup K_4$ in $\mathbb{R}^3$ such that

(i) every three-component sublink of L is the Borromean rings, and

(ii) L is carried onto itself by a group of rotations isomorphic to the rotation group of a regular tetrahedron, acting faithfully on the components.

Our main result is that such a link exists, together with an explicit model (Theorem 7.1). Two features of the problem make it more rigid than it first appears, and both are proved below: condition (i) already forces every pair of components to be unlinked (Lemma 3.1), and condition (ii) already forces the symmetry to permute the four components transitively (Lemma 3.2). So the problem really has two hypotheses rather than the four one might be tempted to impose.

The combinatorial idea the construction rests on is due to Kyle Miller [6]: place one component on each *face* of a tetrahedron. A tetrahedron has four faces and four vertices, and the three faces meeting at a vertex are exactly the three that avoid the opposite face. Choosing three of the four components is therefore the same as naming a vertex, and condition (i) becomes the statement that the components weave in the Borromean pattern around each of the four vertices. Since the symmetry group is transitive on vertices, the four instances of condition (i) are a single condition. This is what makes the problem tractable.

Sections 2–4 set up the problem and the equivariant frame. Section 5 checks that the symmetry is compatible with the linking that condition (i) demands, which is a necessary condition for the problem to have any solution at all. Section 6 proves two obstructions, which together say that the components can be neither flat nor economical. Section 7 gives the link.

2 Conventions

A *link* of n components is a smooth embedding of n disjoint circles in $\mathbb{R}^3$; we write $L = K_1 \sqcup \cdots \sqcup K_n$ and call the K_i its *components*. Two links are *equivalent* if some ambient isotopy of $\mathbb{R}^3$ carries one onto the other respecting the labelling of components. For $S \subseteq \{1, \dots, n\}$, the *sublink* L_S is $\bigcup_{i \in S} K_i$. The *unlink* U_n is n components bounding disjoint embedded discs.

We write $\mathcal{B}$ for the *Borromean rings*: the three-component link L6a4 of the standard tables [5], whose standard diagram is Figure 1. It has the two properties quoted above, namely $\mathrm{lk}(K_i, K_j) = 0$ for each pair while $\mathcal{B}$ is not the unlink. The diagram is drawn with circular arcs, but no three *round circles* in space form the Borromean rings: that is a theorem of Freedman and Skora [1], with an elementary proof due to Lindström and Zetterström [4]. Any realisation therefore deforms the circles, a constraint the present problem inherits.

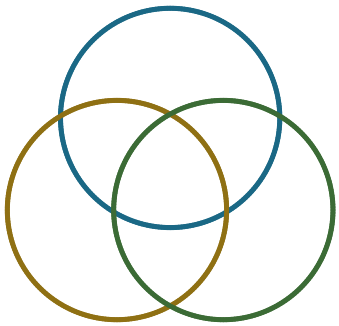


Figure 1: The standard diagram of the Borromean rings $\mathcal{B}$. Each component passes over one neighbour and under the other, cyclically. No two of the three are linked, yet all three together cannot be separated. Here and throughout, over and under are shown only by the break in the lower strand; the colours distinguish components and carry no other information.

The *linking number* $\mathrm{lk}(K_i, K_j) \in \mathbb{Z}$ of two oriented components is half the sum of the signs of the crossings between them in any diagram; it is an invariant of the link, and it vanishes for every pair of $\mathcal{B}$. Where a finer measure is needed we use Milnor's triple invariant. If all the pairwise linking numbers of a link vanish, then for distinct i, j, k the invariant $\bar{\mu}(ijk) \in \mathbb{Z}$ of [7, 8] is defined with no indeterminacy, it is unchanged by an orientation-preserving diffeomorphism of $\mathbb{R}^3$, and it is alternating: interchanging two of its three indices reverses its sign. For the Borromean rings $\bar{\mu}(123) = \pm 1$, the sign depending on the chosen orientations and diagram.

Finally, let $\mathrm{T} \leq SO(3)$ denote the rotation group of a regular tetrahedron: the identity, eight rotations through $\pm 120°$ about the four axes joining a vertex to the centre of the opposite face, and three rotations through $180°$ about the axes joining midpoints of opposite edges, so $|\mathrm{T}| = 12$ and $\mathrm{T} \cong A_4$. An embedded link $L \subset \mathbb{R}^3$ is T-*symmetric* if $g(L) = L$ as a set for every $g \in \mathrm{T}$; this induces a homomorphism $\rho\colon \mathrm{T} \to S_n$ by $g(K_i) = K_{\rho(g)(i)}$. Note that symmetry is a property of an embedding and not of an equivalence class: a link type "has tetrahedral symmetry" when some representative embedding does.

3 The problem, and two things one need not assume

Lemma 3.1. *If every three-component sublink of L is $\mathcal{B}$, then $\mathrm{lk}(K_i, K_j) = 0$ for every pair, and indeed every two-component sublink of L is the unlink U_2.*

Proof. Any two-element subset of $\{1, 2, 3, 4\}$ is contained in a three-element subset S. By hypothesis $L_S \cong \mathcal{B}$, and every two-component sublink of $\mathcal{B}$ is U_2. Sublinks of equivalent links are equivalent, so the two-component sublinks of L are each U_2. □

Lemma 3.2. *Let L have four components and let $\rho\colon \mathrm{T} \to S_4$ be injective. Then $\rho(\mathrm{T}) = A_4$, the action on components is transitive, and the stabiliser of each component is cyclic of order 3, generated by a rotation through 120° about one of the four vertex-to-face axes.*

Proof. $\rho(\mathrm{T})$ is a subgroup of S_4 isomorphic to A_4, hence of order 12 and of index 2; the only such subgroup of S_4 is A_4 itself. The action of A_4 on four points is transitive, so by orbit–stabiliser each stabiliser has order 3. The elements of order 3 in T are precisely the rotations through $\pm 120^\circ$ about the four vertex-to-face axes. □

So Problem 1.1 needs only the two conditions stated. In particular "pairwise unlinked" is not an independent requirement, and transitivity on components is a consequence rather than a hypothesis.

4 The tetrahedral frame

Fix the regular tetrahedron Δ with vertices

$$v_1 = (1,1,1), \quad v_2 = (1,-1,-1), \quad v_3 = (-1,1,-1), \quad v_4 = (-1,-1,1),$$

and let F_i denote the face of Δ opposite v_i. By Lemma 3.2 we may label the components so that the stabiliser of K_i is generated by the rotation r_i through 120° about the axis through v_i; we then say K_i *belongs to* F_i. In these coordinates r_1 is the cyclic permutation of coordinates

$$r_1(x,y,z) = (z,x,y), \tag{1}$$

and the three half-turns about the coordinate axes,

$$u_2(x,y,z) = (x,-y,-z), \qquad u_3(x,y,z) = (-x,y,-z), \qquad u_4(x,y,z) = (-x,-y,z), \tag{2}$$

satisfy $u_i(v_1) = v_i$ and so represent the three nontrivial cosets of $\langle r_1 \rangle$ in T.

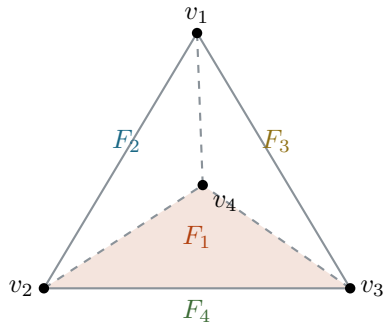
Lemma 4.1 (Generation principle)**.** *Let $K \subset \mathbb{R}^3$ be any closed curve with $r_1(K) = K$, and set $K_1 = K$, $K_i = u_i(K)$ for $i = 2,3,4$. If these four curves are pairwise disjoint, then $L = K_1 \sqcup K_2 \sqcup K_3 \sqcup K_4$ is* T*-symmetric with ρ injective.*

Proof. Write $u_1 = \mathrm{id}$. For $g \in \mathrm{T}$ and $i \in \{1,2,3,4\}$ there are a unique j and a unique $m \in \{0,1,2\}$ with $g\,u_i = u_j r_1^m$, because the u_i represent the cosets of $\langle r_1 \rangle$. Hence $g(K_i) = g\,u_i(K) = u_j r_1^m(K) = u_j(K) = K_j$, so g permutes the components. The resulting permutation determines g on cosets, and since T acts faithfully on $\mathrm{T}/\langle r_1 \rangle$, ρ is injective. □

Lemma 4.1 is what makes the symmetry exact rather than approximate: one designs a single r_1-invariant curve and the group produces the rest, so there is no step at which the symmetry could be lost.

Lemma 4.2 (Vertices and triples)**.** *The map $S \mapsto v_k$, where $\{k\} = \{1,2,3,4\} \setminus S$, is a bijection from three-element subsets of $\{1,2,3,4\}$ to vertices of Δ, and it is equivariant for the actions of* T*. The components belonging to the three faces meeting at v_k are exactly $\{K_i : i \neq k\}$.*

Proof. The three faces through v_k are those containing v_k, namely F_i for $i \neq k$, since F_k is the face opposite v_k. Complementation is a bijection and commutes with the induced action. □



drop component	weave at vertex
K_1	v_1
K_2	v_2
K_3	v_3
K_4	v_4

Figure 2: One component per face. Component K_i belongs to the face F_i opposite the vertex v_i. Deleting K_k leaves the three components belonging to the three faces that meet at v_k, so the four instances of condition (i) correspond to the four vertices, and the symmetry group permutes them transitively.

5 The symmetry is compatible with the required linking

Before constructing anything it is worth checking that conditions (i) and (ii) are not in conflict, since a symmetry constrains the invariants of a link and could in principle force them to vanish. They are not in conflict, and the reason the check is clean is the following.

Lemma 5.1. *The components of a* T*-symmetric four-component link may be oriented so that every* $g \in \mathrm{T}$ *preserves the orientation of every component it maps to itself, and carries chosen orientations to chosen orientations throughout.*

Proof. An equivariant choice of orientations is a section of the associated $\{\pm 1\}$-bundle over the component set $\mathrm{T}/\langle r_1 \rangle$, and such bundles are classified by $\mathrm{Hom}(\langle r_1 \rangle, \{\pm 1\})$. Since $\langle r_1 \rangle$ has order 3, which is odd, that group is trivial, so the bundle is trivial and a section exists. □

With orientations so chosen, every $g \in \mathrm{T}$ is an orientation-preserving diffeomorphism of $\mathbb{R}^3$ carrying oriented components to oriented components, so for $\sigma = \rho(g)$,

$$\bar{\mu}(\sigma i, \sigma j, \sigma k) = \bar{\mu}(i, j, k). \tag{3}$$

Proposition 5.2. *Let* L *satisfy (ii), with all pairwise linking numbers zero. Then the vector*

$$\big(\bar{\mu}(234),\ \bar{\mu}(134),\ \bar{\mu}(124),\ \bar{\mu}(123)\big)$$

lies in a one-dimensional subspace of $\mathbb{R}^4$*, spanned by* $(-1, 1, -1, 1)$*. In particular condition (i), which forces* $|\bar{\mu}(ijk)| = 1$ *on all four triples, is compatible with (ii), and it determines the four signs up to a single global sign.*

Proof. Since $\bar{\mu}$ is alternating it is determined by its values on the four three-element subsets written in increasing order, which are the four coordinates above. Equation (3) for σ ranging over $\rho(\mathrm{T}) = A_4$ is a system of linear equations in those four unknowns. Take $\sigma = (123)$. Applying (3) to the triples $(1, 2, 4)$, $(2, 3, 4)$ and $(3, 1, 4)$ in turn, and using that $\bar{\mu}$ is alternating to re-sort indices, gives $\bar{\mu}(234) = \bar{\mu}(124)$ and $\bar{\mu}(134) = -\bar{\mu}(124)$. Taking $\sigma = (234)$ and arguing the same way gives $\bar{\mu}(123) = -\bar{\mu}(124)$. Since (123) and (234) generate A_4, these are all the relations, and the solution space is spanned by $(1, -1, 1, -1)$ in the ordering above, equivalently by $(-1, 1, -1, 1)$. □

6 Two obstructions

Both obstructions in this section say that an obvious economy is unavailable. They are what force the construction of Section 7 to look as elaborate as it does.

6.1 The components cannot be flat and convex

Call a closed curve *convex planar* if it lies in a plane and bounds a convex region there; equivalently, no straight line meets it in more than two points. Triangles and ellipses are convex planar; a star-shaped polygon is not.

Theorem 6.1. *No* T*-symmetric four-component link all of whose components are convex planar satisfies condition (i).*

Proof. Let a_i be the axis through v_i, so that K_i is invariant under the rotation r_i of order 3 about a_i.

Step 1: the planes are forced. A planar simple closed curve spans a unique plane P_i, and $r_i(K_i) = K_i$ forces $r_i(P_i) = P_i$. A rotation through $120°$ about a_i carries a plane containing a_i to a different such plane, since $120° \not\equiv 0 \pmod{180°}$; hence $P_i \perp a_i$.

Step 2: the half-turn reverses the line of intersection. For $i \neq j$ the axes a_i, a_j meet at an angle $\arccos(-1/3)$, so P_i and P_j are not parallel and $\ell = P_i \cap P_j$ is a line. There is a unique element $u \in \mathrm{T}$ of order 2 with $u(K_i) = K_j$ and $u(K_j) = K_i$, namely the half-turn realising the double transposition $(ij)(kl)$; let b be its axis. Then $u(\ell) = \ell$. A half-turn carries a line to itself only if the line is b, or meets b orthogonally. The first is impossible, as b lies in neither P_i nor P_j; in the second case $u|_\ell$ is the point reflection of ℓ in the point $\ell \cap b$. Concretely, for $i = 1$, $j = 2$ we have b the x-axis, and with $P_1 = \{x + y + z = c\}$ and $P_2 = \{x - y - z = c\}$ one computes $\ell = \{(c, s, -s) : s \in \mathbb{R}\}$, whose direction $(0, 1, -1)$ is orthogonal to b and which meets b at $(c, 0, 0)$.

Step 3: chords cannot nest. Being convex, K_i meets ℓ in at most two points, so (generically) $K_i \cap \ell$ is the pair of endpoints of an interval $I_i \subset \ell$, and likewise for K_j. Two disjoint convex planar curves lying in distinct planes are linked precisely when I_i and I_j interleave, and unlinked precisely when the intervals nest or are disjoint. By Step 2, I_j is the image of I_i under the point reflection in $m = \ell \cap b$. Taking m as the origin of a coordinate on ℓ and writing $I_i = [\alpha, \beta]$, we get $I_j = [-\beta, -\alpha]$. For $I_i \subseteq I_j$ we would need $-\beta \leq \alpha$ and $\beta \leq -\alpha$, that is $\alpha + \beta \geq 0$ and $\alpha + \beta \leq 0$; so $\alpha + \beta = 0$, $I_i = I_j$, and the two curves meet. Nesting is therefore unavailable (Figure 3).

Step 4: neither survivor works. The group A_4 is 2-transitive on four points, hence transitive on the six unordered pairs, so every pair falls into the same one of the remaining two cases. If all pairs interleave then $\mathrm{lk}(K_i, K_j) = \pm 1$ for every pair, contradicting Lemma 3.1. If all pairs are disjoint, then the convex disc $D_i \subset P_i$ bounded by K_i meets P_j only in the filled interval I_i, which is disjoint from K_j; so D_i is a disc bounded by K_i and disjoint from every other component, every three-component sublink is split, and no such sublink is $\mathcal{B}$. □

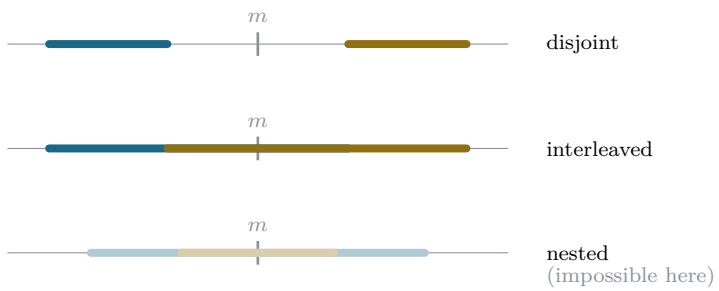


Figure 3: The three ways two intervals can sit on the line $\ell = P_i \cap P_j$. The half-turn exchanging the two components reflects ℓ in the point m, so one interval is the mirror image of the other through m. Disjoint and interleaved configurations are compatible with that; nesting is not, and nesting is the only one of the three that yields a Borromean triple.

6.2 The components cannot cross the minimum number of times

Theorem 6.1 forces the components off their planes. The next result forces them to cross each other more often than one might hope.

Suppose, as the picture in Figure 2 suggests, that each component runs along the boundary of its face. Two faces F_i, F_j share an edge whose endpoints are the two vertices v_k, v_l with $\{k,l\} = \{1,2,3,4\} \setminus \{i,j\}$, so the components K_i and K_j come close together near v_k and near v_l and nowhere else. The most economical diagram consistent with that picture is one in which the three components at each vertex cross each other once apiece, giving $4 \times 3 = 12$ crossings in all and exactly two crossings per pair — the fewest any link satisfying our conditions could have, since a pair crossing fewer than twice has nonzero linking number or does not interact at all.

Theorem 6.2. *There is no* T*-symmetric diagram of a four-component link in which, for each vertex v_k, the three components $\{K_i : i \neq k\}$ pairwise cross exactly once near v_k, and no crossings occur elsewhere.*

Proof. Fix such a diagram. Near v_k the three components cross pairwise once, so the relation "K_i passes over K_j near v_k" is a tournament on three elements. The stabiliser of v_k in T is $\langle r_k \rangle$, which permutes those three components in a 3-cycle and preserves the relation; a 3-cycle preserves no linear order on three elements, so the relation is a cyclic order, and there are exactly two of them. Since T is transitive on the vertices of Δ, the choice at one vertex determines the choice at all four, so there are exactly two diagrams to consider, interchanged by reversing every cyclic order.

Each pair $\{K_i, K_j\}$ crosses exactly twice, at v_k and at v_l. Two crossings between distinct components contribute ± 1 each to $2\,\mathrm{lk}(K_i, K_j)$, and the two contributions cancel exactly when the *same* component passes over at both crossings. By Lemma 3.1 we therefore need the over-strand of the pair $\{K_i, K_j\}$ to be the same at v_k and at v_l.

Take the cyclic order at v_k to be the orbit of r_k, the rotation through $+120°$ about the axis through v_k. In the coordinates of (1)–(2) these rotations act on indices as

$$r_1 = (2\,3\,4), \qquad r_2 = (1\,4\,3), \qquad r_3 = (1\,2\,4), \qquad r_4 = (1\,3\,2),$$

and the over-strand of a pair at a vertex is read off from the corresponding cyclic order. The six pairs give

pair	vertices	over at the first	over at the second
$\{K_1, K_2\}$	v_3, v_4	K_1	K_2
$\{K_1, K_3\}$	v_2, v_4	K_3	K_1
$\{K_1, K_4\}$	v_2, v_3	K_1	K_4
$\{K_2, K_3\}$	v_1, v_4	K_2	K_3
$\{K_2, K_4\}$	v_1, v_3	K_4	K_2
$\{K_3, K_4\}$	v_1, v_2	K_3	K_4

In every row the over-strand changes, so every pair has linking number ± 1. Reversing all four cyclic orders exchanges the two columns and changes nothing. Both diagrams are therefore excluded by Lemma 3.1. □

Theorems 6.1 and 6.2 together explain the shape of what follows: the components must leave their planes, and they must meet each other more than twice per pair.

7 The link

Theorem 7.1. *Let*

$$p = (3, -1, -1), \qquad q = (-1, 0, -2),$$

let K_1 be the closed hexagon through

$$p, \quad q, \quad r_1(p), \quad r_1(q), \quad r_1^2(p), \quad r_1^2(q)$$

in that cyclic order, and let $K_i = u_i(K_1)$ for $i = 2, 3, 4$ as in (2). Then $L = K_1 \sqcup K_2 \sqcup K_3 \sqcup K_4$ is a four-component link with all twenty-four vertices in $\mathbb{Z}^3$, it is T*-symmetric with ρ injective, all six of its pairwise linking numbers vanish, and each of its four three-component sublinks is the Borromean rings.*

Written out, with $r_1(x, y, z) = (z, x, y)$, the four hexagons are the rows of Table 1; Figure 4 shows the link along two of its symmetry axes.

component	vertices
K_1	$(3,-1,-1)$ $(-1,0,-2)$ $(-1,3,-1)$ $(-2,-1,0)$ $(-1,-1,3)$ $(0,-2,-1)$
K_2	$(3,1,1)$ $(-1,0,2)$ $(-1,-3,1)$ $(-2,1,0)$ $(-1,1,-3)$ $(0,2,1)$
K_3	$(-3,-1,1)$ $(1,0,2)$ $(1,3,1)$ $(2,-1,0)$ $(1,-1,-3)$ $(0,-2,1)$
K_4	$(-3,1,-1)$ $(1,0,-2)$ $(1,-3,-1)$ $(2,1,0)$ $(1,1,3)$ $(0,2,-1)$

Table 1: The twenty-four vertices of L. Each row is a closed hexagon, listed in cyclic order; K_2, K_3, K_4 are the images of K_1 under the half-turns u_2, u_3, u_4 about the three coordinate axes.

Proof of the first three assertions. Every element of T used above is a signed permutation of coordinates, so it maps $\mathbb{Z}^3$ to $\mathbb{Z}^3$ and all twenty-four vertices are integral. The curve K_1 is r_1-invariant by construction, since r_1 shifts the listed cyclic order by two places; the four curves are pairwise disjoint, the closest approach between two of them being $0.3123\ldots$ against a diameter of $6.63\ldots$. So Lemma 4.1 applies and gives T-symmetry with ρ injective.

For the linking numbers, choose any direction of projection in which the image is a diagram, compute the finitely many crossings between distinct components, and sum their signs; all the arithmetic is exact because the vertices are integral. Each of the six sums is zero. □

The remaining assertion is the identification of the sublinks, and here our argument is computational.

Proposition 7.2. *Each three-component sublink of the link L of Theorem 7.1 has all three pairwise linking numbers zero and has the same Jones polynomial [2, 3] as $\mathcal{B}$. Consequently each is $\mathcal{B}$.*

Verification. By Lemma 4.2 the four sublinks form a single orbit under T, so they are equivalent to one another and it suffices to treat one; we verified all four independently as a check. For each, a projection direction was chosen by search, the resulting diagram was simplified using Reidemeister moves of the first and second kinds, and the Kauffman bracket of the reduced diagram was expanded and normalised by the writhe. Each sublink projects to an 18-crossing diagram that reduces to 8 crossings, with the Jones polynomial

$$-A^{-12} + 3A^{-8} - 2A^{-4} + 4 - 2A^4 + 3A^8 - A^{12},$$

in Kauffman's variable A, which is the Jones polynomial of $\mathcal{B}$; and the three pairwise linking numbers are zero. Among three-component links of small crossing number, this combination of invariants occurs only for $\mathcal{B}$ [5]. The computations are deterministic and are reproduced by the scripts accompanying this paper. □

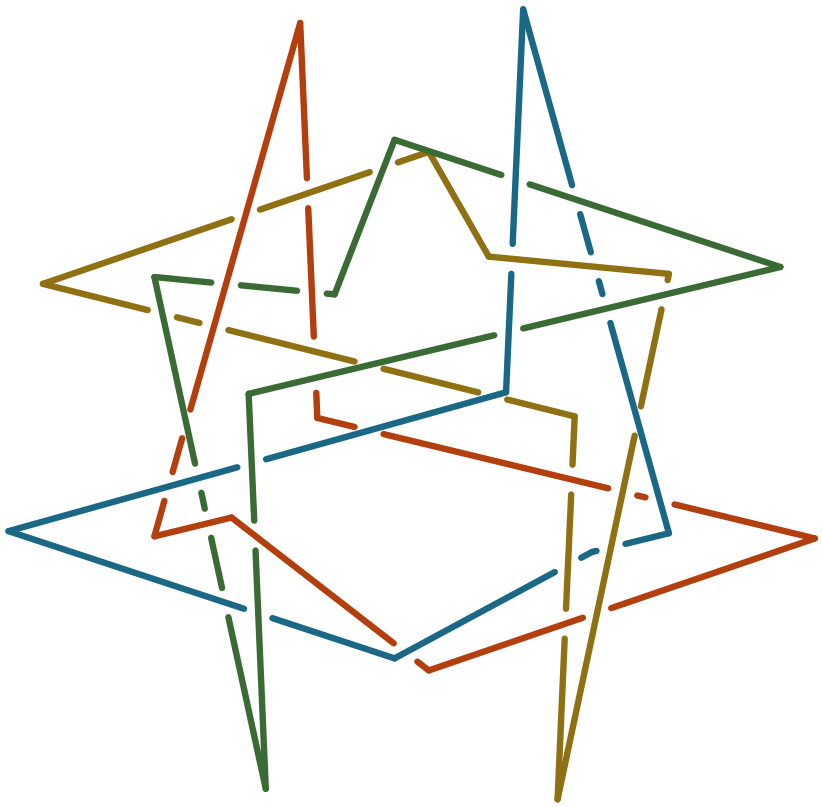


Figure 4: The link L of Theorem 7.1. The projection direction was chosen among those giving the fewest crossings to keep the crossings as far apart from one another, and from the corners, as possible. Components are coloured K_1, K_2, K_3, K_4.

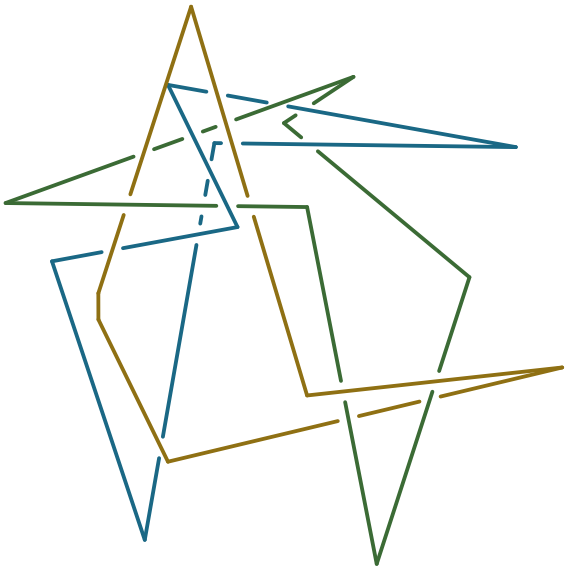


Figure 5: The sublink obtained by deleting K_1. By Lemma 4.2 the three surviving components are those belonging to the faces meeting at v_1, and by Proposition 7.2 they form the Borromean rings. The other three sublinks are carried onto this one by the symmetry.

8 Remarks

What is and is not proved. The symmetry in Theorem 7.1 is exact and is established by Lemma 4.1 rather than measured; the vanishing of the pairwise linking numbers is a finite exact computation. The identification of the sublinks as $\mathcal{B}$, however, rests on Proposition 7.2, which appeals to the tabulation of links of small crossing number. A synthetic argument — an explicit isotopy carrying one sublink onto the standard diagram of $\mathcal{B}$, which by Lemma 4.2 would settle all four at once — would be preferable, and we do not have one. The natural route is to redesign the component so that the Borromean weave is visible in a neighbourhood of each vertex and the connecting arcs demonstrably meet nothing.

Economy. Theorem 6.2 rules out 12 crossings. The link of Theorem 7.1 is far from that bound: its best projection among 192 sampled directions has 36 crossings. We make no claim about the minimal crossing number of this link, nor about whether the link of Theorem 7.1 is the simplest one satisfying Problem 1.1.

Acknowledgement. The observation that the four components should be indexed by the faces of a tetrahedron, so that triples of components correspond to vertices, is due to Kyle Miller [6].

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